W07

Making maps: QGIS, ArcGIS Pro and tmap

GIS & MAPPING

A good map explains itself: it has a clear purpose, a visual hierarchy and just the elements it needs. Today you make the same study-area map in QGIS, in ArcGIS Pro and with tmap in R, and decide which tool suits each situation.

DURATION	About 2 hours.
YOU NEED	QGIS, R with `tmap` (v4), `sf` and `terra`; ArcGIS Pro optional.
FILES	`week07_maps_tmap.R`, `week07_solutions.R`, the Week 5 data plus `bird_counts_wide.csv`, `species_traits.csv`
DATA	Sierra de Montfred (fictitious).

Learning outcomes

- design a study-area map and a results map for publication;
- build a QGIS print layout with an overview map, scale bar, north arrow and legend;
- do the same in ArcGIS Pro;
- make reproducible maps with tmap, including small multiples (facets);
- choose colour palettes suited to each type of data.

1 · Principles

- **One message per map.** A study-area map locates; a results map shows a pattern. Do not try to do both.
- **Palette by data type:** sequential for ordered values (elevation), diverging when there is a meaningful centre (positive/negative change), qualitative for categories (land cover).
- **Accessibility:** avoid red versus green as the only contrast; viridis and ColorBrewer palettes work for most colour-blind readers.
- **Just enough elements:** always a scale bar; a north arrow only if north is not up or the map is for fieldwork; a legend with units.

2 · Print layout in QGIS

1. Open week05.qgz and leave the hillshade, DEM, park, streams and points visible.
2. **Project > New Print Layout**; name it "Study area". In the layout, right-click the page > **Page Properties**: A4 landscape.
3. **Add Item > Add Map** and draw a rectangle. In **Item Properties** set the scale (for example 1:60,000) and tick *Lock layers*.
4. Overview map: add a second small map in a corner with a basemap. In its properties, **Overviews > +** and choose *Map 1*: a frame shows the study area.
5. **Add Item** > **Scale Bar**, **North Arrow** and **Legend**. In the legend, untick *Auto update* and remove layers you do not need.
6. Optional: a **Grid** in the map properties, with UTM coordinates on the frame.
7. **Layout > Export as Image** (300 dpi) or **Export as PDF** (vector).

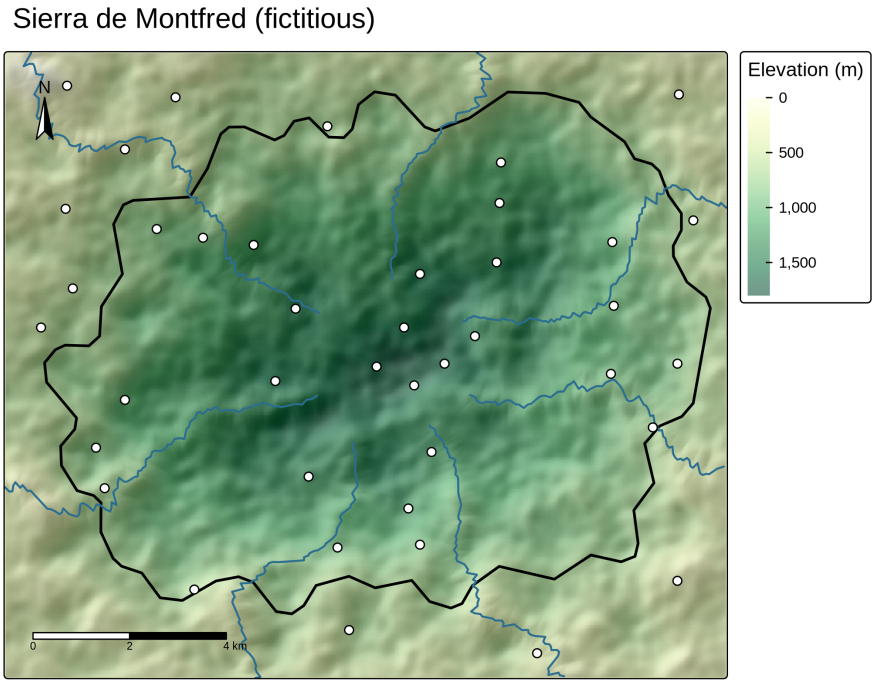
ArcGIS Pro

1. **Insert > New Layout** > A4 landscape.
2. **Insert > Map Frame** and draw it. Set the scale in the bottom bar.
3. **Insert > North Arrow**, **Scale Bar** and **Legend**. For the overview, insert another map frame and add an **Extent Indicator**.
4. **Share > Export Layout**: PNG at 300 dpi or PDF.

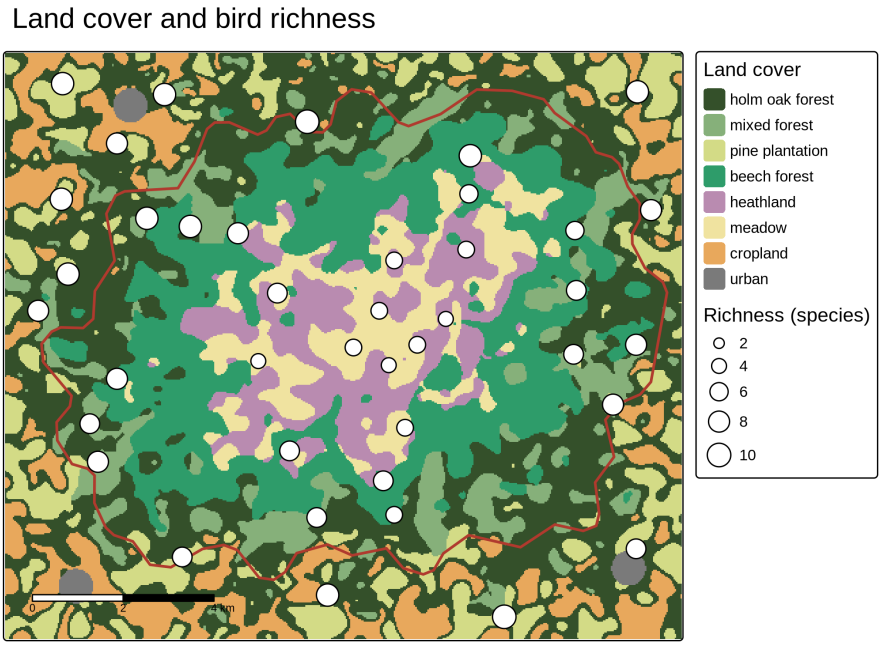
3 · Reproducible maps with tmap

tmap works like ggplot2: each `tm_shape()` adds some data and the functions that follow say how to draw it. The script uses tmap 4 syntax: colours are set with `col.scale` / `fill.scale` and legends with `tm_legend()`.

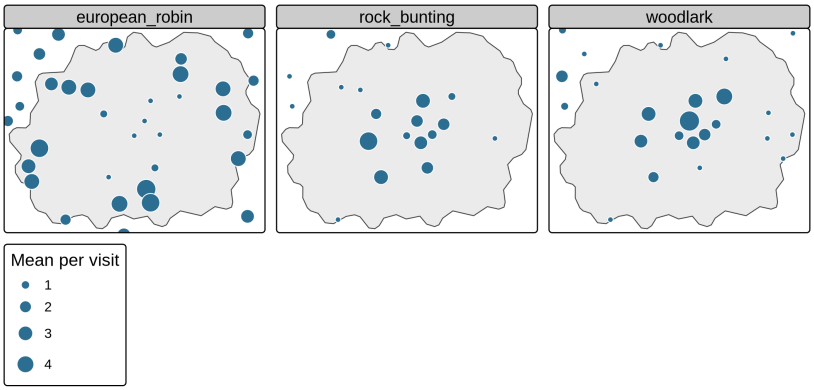
```
tm_shape(hill) + tm_raster(col.scale = tm_scale_continuous(values = "brewer.greys"),
                           col.legend = tm_legend_hide()) +
  tm_shape(dem) + tm_raster(col_alpha = 0.55,
                            col.legend = tm_legend(title = "Elevation (m)")) +
  tm_shape(park) + tm_borders(lwd = 2) +
  tm_shape(pts) + tm_symbols(shape = 21, fill = "white", size = 0.45) +
  tm_scalebar(breaks = c(0, 2, 4)) + tm_compass()
```



Study-area map with tmap: hillshade, elevation, park boundary, streams and points.



Results map: land cover (qualitative palette) and bird richness at each point (circle size).



Small multiples with `tm_facets()`*: same scale and legend, one species per panel.*

Export with `tmap_save(map, "map.png", width = 18, height = 14, units = "cm", dpi = 300)`. `tmap_mode("view")` turns the same code into an interactive web map for exploring the data.

Which tool should I use?

QGIS / ArcGIS Pro: one-off maps with careful design (posters, reports). **tmap / ggplot2:** maps you redo when the data change, map series and paper figures. Many ecologists prepare data in R and finish the final map in QGIS.

Exercises

Answers in `week07_solutions.R`.

1. Change the DEM palette to `hcl.terrain2` and remove the hillshade. Which version reads better?
2. Add stream names to the land-cover map (`tm_text()`).
3. Slope map in intervals (0–5, 5–15, 15–25, > 25°) with `tm_scale_intervals()`.
4. Redo the small multiples with three other species.
5. Export the results map as a 17 cm wide PDF.
6. **QGIS:** finish your study-area layout and export it at 300 dpi.

Before Week 8

- Create a free account at gbif.org.
- Install `install.packages(c("rgbif", "CoordinateCleaner"))`.